Statistical Inference for Linear Stochastic Approximation with Richardson-Romberg Extrapolation

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Abstract

Your abstract.

1. Introduction

Stochastic approximation (SA) algorithms are a cornerstone of modern computational statistics, optimization, and reinforcement learning. Introduced by Robbins and Monro (1951), these iterative procedures provide a principled way to find roots of equations or optimize objectives when only noisy observations are available. A particularly important subclass is the *linear stochastic approximation* (LSA) algorithm, which arises naturally in temporal-difference (TD) learning, policy evaluation, and stochastic gradient descent for linear models.

In this work, we study the LSA recursion with a *constant step size* $\alpha > 0$:

$$\theta_k^{(\alpha)} = \theta_{k-1}^{(\alpha)} - \alpha\Big\{A(Z_k)\theta_{k-1}^{(\alpha)} - b(Z_k)\Big\}, \quad k \geq 1,$$

where $\{Z_k\}_{k\in\mathbb{N}}$ is a time-homogeneous Markov chain on a measurable space $(\mathsf{Z}, \mathcal{Z})$ with transition kernel Q and unique invariant distribution π. The mappings $A : \mathsf{Z} \to \mathbb{R}^{d\times d}$ and $b : \mathsf{Z} \to \mathbb{R}^d$ are measurable functions satisfying $\overline{A} := \int_{\mathsf{Z}} A(z)d\pi(z)$ and $\overline{b} := \int_{\mathsf{Z}} b(z)d\pi(z)$. Assuming that $-\overline{A}$ is a *Hurwitz matrix* (all eigenvalues have strictly negative real parts), the target parameter $\theta^* = \overline{A}^{-1}\overline{b}$ is uniquely defined.

The *Polyak–Ruppert averaging* procedure (Polyak, 1990; Ruppert, 1988) provides an effective variance reduction technique. Given a burn-in period $n_0 \geq 0$, the averaged iterate is defined as

$$\overline{\theta}_n^{(\alpha)} = \frac{1}{n - n_0}\sum_{k=n_0}^{n-1}\theta_k^{(\alpha)}.$$

1.1. Bias of constant step-size iterates

The use of a constant step size $\alpha > 0$ offers several practical advantages: it enables geometrically fast forgetting of the initial condition (Dieuleveut, Durmus, and Bach, 2020) and simplifies hyperparameter tuning compared to diminishing step-size schedules. However, unlike the classical regime $\alpha_k \to 0$ with $\sum \alpha_k = \infty$ and $\sum \alpha_k^2 < \infty$, a constant step size produces iterates that converge only *in distribution* to a stationary measure Π_α, rather than almost surely to θ^*. The stationary expectation $\mathbb{E}\big[\theta_\infty^{(\alpha)}\big]$ is generally *biased* with respect to θ^*, and this bias cannot be eliminated by Polyak–Ruppert averaging alone.

As shown in Levin, Naumov, and Samsonov (2025) and Huo, Chen, and Xie (2023), the asymptotic bias admits a power-series expansion in α:

$$\lim_{n\to\infty}\mathbb{E}\big[\theta_n^{(\alpha)}\big] = \theta^* + \alpha\Delta + O(\alpha^2),$$

where $\Delta = \overline{A}^{-1} \sum_{k=1}^{\infty} \mathbb{E}\left[\left\{Q^k \tilde{A}(Z_\infty)\right\}\epsilon(Z_\infty)\right]$ depends on the correlation structure of the Markov chain, and $\tilde{A}(z) = A(z) - \overline{A}$ is the centered matrix-valued function.

1.2. Richardson–Romberg extrapolation

To eliminate the leading $O(\alpha)$ bias term, we employ the *Richardson–Romberg* (RR) *extrapolation* procedure. Two LSA sequences are run *on the same Markov chain trajectory* $\{Z_k\}$ with step sizes α and 2α, and the RR iterate is formed as

$$\overline{\theta}_n^{(\alpha,\,\mathrm{RR})} = 2\overline{\theta}_n^{(\alpha)} - \overline{\theta}_n^{(2\alpha)}.$$

Since both sequences share the same noise realization, the leading bias term $\alpha\Delta$ cancels, leaving a residual bias of order $O\big(\alpha^{3/2}\big)$ or higher (Levin et al., 2025).

More generally, one can consider the multi-level extrapolation with M step sizes $\mathcal{A} = \{\alpha_1, ..., \alpha_M\}$ and coefficients $\{h_m\}$ determined by the Vandermonde system (Huo et al., 2023):

$$\sum_{m=1}^{M} h_m = 1, \quad \sum_{m=1}^{M} h_m \alpha_m^l = 0, \quad l = 1, ..., M-1,$$

which reduces the bias from $O(\alpha)$ to $O(\alpha^M)$.

1.3. Problem statement and goals

The high-order moment bounds for the RR iterate $\overline{\theta}_n^{(\alpha,\,\mathrm{RR})}$ established in Levin et al. (2025) show that the leading error term scales as $\sqrt{\mathrm{Tr}\ \Sigma_\epsilon^{(\mathrm{M})}} \cdot n^{-1/2}$, matching the minimax optimal rate. Berry–Esseen type bounds and bootstrap inference procedures for the *standard* Polyak–Ruppert average $\overline{\theta}_n$ (without extrapolation) under Markovian noise have been obtained in Samsonov, Sheshukova, Moulines, and Naumov (2025).

However, the *distributional approximation* — specifically, the central limit theorem and Berry–Esseen bounds — for the *averaged Richardson–Romberg iterates* $\overline{\theta}_n^{(\alpha,\,\mathrm{RR})}$ remains an open problem. The main goal of this work is to fill this gap by establishing:

1. A *central limit theorem* for $\sqrt{n}\big(\overline{\theta}_n^{(\alpha,\,\mathrm{RR})} - \theta^*\big)$, identifying the limiting covariance matrix.
2. *Non-asymptotic Berry–Esseen bounds* for the rate of convergence to the Gaussian limit.
3. A practical *inference procedure* (confidence intervals) based on the RR-extrapolated iterates.

1.4. Setting and assumptions

We now formalize the setting and state the assumptions that will be used throughout this work.

Let $\{Z_k\}_{k\in\mathbb{N}}$ be a Markov chain on a complete separable metric space $(Z, \mathcal{Z})$ with transition kernel Q.

Assumption 1 (Uniform geometric ergodicity). The kernel Q admits a unique invariant distribution π and is *uniformly geometrically ergodic*: there exists $t_{\mathrm{mix}} \in \mathbb{N}^*$ such that for all $k \in \mathbb{N}^*$,

$$\Delta\big(Q^k\big) \coloneqq \sup_{z,z'\in Z} \frac{1}{2}\, \|Q^k(z,\cdot) - Q^k(z',\cdot)\|_{\mathrm{TV}} \le (1/4)^{\lfloor k/t_{\mathrm{mix}}\rfloor}.$$

Equivalently, there exist constants $\zeta > 0$ and $\rho \in (0,1)$ such that $\sup_z \|Q^k(z,\cdot) - \pi\|_{\mathrm{TV}} \le \zeta\rho^k$ for all $k \ge 1$.

Assumption 2 (Hurwitz condition and boundedness). The matrix $-\overline{A}$ is Hurwitz, i.e., all eigenvalues of $\overline{A}$ have strictly positive real parts. Moreover,

$$C_A \coloneqq \sup_{z\in Z}\|A(z)\| \vee \sup_{z\in Z}\|\tilde{A}(z)\| < \infty,$$

where $\tilde{A}(z) = A(z) - \overline{A}$.

Assumption 3 (Noise regularity). The noise function $\epsilon(z) = \tilde{A}(z)\theta^* - \tilde{b}(z)$, where $\tilde{b}(z) = b(z) - \overline{b}$, satisfies

$$\|\epsilon\|_\infty := \sup_{z\in Z}\|\epsilon(z)\| < +\infty.$$

Under Assumptions 1–3, the error $\theta_k^{(\alpha)} - \theta^*$ satisfies the recursion

$$\theta_k^{(\alpha)} - \theta^* = (I - \alpha A(Z_k))\left(\theta_{k-1}^{(\alpha)} - \theta^*\right) - \alpha\epsilon(Z_k).$$

1.5. Key quantities

The *Markovian noise covariance matrix* captures both the marginal variance and the temporal correlations of the noise:

$$\Sigma_\epsilon^{(\mathrm{M})} = \mathbb{E}_\pi\left[\epsilon(Z_0)\epsilon(Z_0)^\top\right] + 2\sum_{\ell=1}^{\infty}\mathbb{E}_\pi\left[\epsilon(Z_0)\epsilon(Z_\ell)^\top\right].$$

This matrix is the limiting covariance in the Markov chain CLT for the partial sums $n^{-1/2}\sum_{t=0}^{n-1}\epsilon(Z_t)$ (cf. Douc et al., 2018, Theorem 21.2.10).

The *asymptotically optimal covariance matrix* is given by

$$\Sigma_\infty = \overline{A}^{-1}\Sigma_\epsilon^{(\mathrm{M})}\left(\overline{A}^{-1}\right)^\top.$$

This covariance is optimal in the Hájek–Le Cam sense and represents the best achievable asymptotic variance for estimating θ^* via averaged LSA.

The *Lyapunov equation* plays a central role in the contraction analysis. For any $P = P^\top \succ 0$, there exists a unique $Q = Q^\top \succ 0$ satisfying $\overline{A}^\top Q + Q\overline{A} = P$. Defining $a = \lambda_{\min}(P)/(2\,\|Q\|)$ and $\kappa_Q = \lambda_{\max}(Q)/\lambda_{\min}(Q)$, the key contraction property holds: for all $\alpha \in [0, \alpha_\infty]$,

$$\|I - \alpha\overline{A}\|_Q^2 \le 1 - \alpha a.$$

Since the iterates $\theta_k^{(\alpha)}$ alone are generally not Markovian (due to the Markovian noise), we consider the *joint process* $\left(\theta_k^{(\alpha)}, Z_{k+1}\right)$ with kernel

$$\overline{P}_\alpha f(\theta, z) = \int_Z Q(z, dz')f(F_{z'}(\theta), z'),$$

where $F_z(\theta) = (I - \alpha A(z))\theta + \alpha b(z)$. Under Assumptions 1–3, this joint chain admits a unique invariant distribution Π_α for sufficiently small $\alpha > 0$ (Levin et al., 2025).

2. Constant Asymptotic

2.1. Linear Stochastic Approximation (LSA)

We consider the recursion

$$\theta_k = \theta_{k-1} - \alpha_k(A(Z_k)\theta_{k-1} - b(Z_k)), \quad \alpha_k = \alpha = \text{const.}$$

Define the transition products

$$\Gamma_{m:k} = \prod_{l=m}^{k}(I - \alpha A(Z_l)).$$

We also introduce

$$J_k^{(0)} = -\sum_{l=1}^{k} \alpha\, \Gamma_{l+1:k}\, \epsilon(Z_l), \quad H_k^{(0)} = -\sum_{l=1}^{k} \alpha\, \Gamma_{l+1:k}\Big(A(Z_l) - \overline{A}\Big) J_{l-1}^{(0)},$$

with initial conditions $J_0^{(0)} = H_0^{(0)} = 0$.

A standard decomposition yields

$$\sqrt{n}\Big(\theta_n^{(\alpha)} - \theta^*\Big) = W + D_1,$$

where

$$W = \frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l\, \epsilon(Z_l), \quad D_1 = \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \Gamma_{1:k}(\theta_0 - \theta^*) + \frac{1}{\sqrt{n}} \sum_{k=1}^{n-1} J_k^{(0)},$$

and

$$Q_l = \alpha \sum_{k=l}^{n-1} \Big(I - \alpha\overline{A}\Big)^{n-k-1}.$$

2.2. Richardson-Romberg (RR) Step

Consider

$$\begin{aligned}
\theta_n^{(\mathrm{RR},\alpha)} - \theta^* &= 2\Big(\theta_n^{(\alpha)} - \theta^*\Big) - \Big(\theta_n^{(2\alpha)} - \theta^*\Big) \\
&= \Big[2\Gamma_{1:n}^{(\alpha)}(\theta_0 - \theta^*) - \Gamma_{1:n}^{(2\alpha)}(\theta_0 - \theta^*)\Big] \\
&\quad + \Big[2J_n^{(0,\alpha)} - J_n^{(0,2\alpha)}\Big] + \Big[2J_n^{(1,\alpha)} - J_n^{(1,2\alpha)}\Big] + \Big[2H_n^{(1,\alpha)} - H_n^{(1,2\alpha)}\Big].
\end{aligned}$$

Define

$$\tilde{J}_n^{(0,\alpha)} = 2J_n^{(0,\alpha)} - J_n^{(0,2\alpha)}, \quad J_n^{(0,\alpha)} = -\alpha \sum_{j=1}^{n} \Big(I - \alpha\overline{A}\Big)^{n-j} \epsilon\big(Z_j\big).$$

Then

$$
\begin{aligned}
\tilde{J}_n^{(0,\alpha)} &= -2\alpha \sum_{j=1}^{n} \Big[\big(I - \alpha\overline{A}\big)^{n-j} - \big(I - 2\alpha\overline{A}\big)^{n-j} \Big] \epsilon(Z_j) \\
&= -2\alpha^2 \overline{A} \sum_{j=1}^{n} \left[\sum_{i=1}^{n-j} \big(I - \alpha\overline{A}\big)^{i-1} \big(I - 2\alpha\overline{A}\big)^{n-j-i} \right] \epsilon(Z_j) \\
&= -2\alpha^2 \overline{A} \sum_{j=1}^{n} H_j^{(n)} \, \epsilon(Z_j),
\end{aligned}
$$

where

$$
H_j^{(n)} := \sum_{i=1}^{n-j} \big(I - \alpha\overline{A}\big)^{i-1} \big(I - 2\alpha\overline{A}\big)^{n-j-i}.
$$

2.3. Norm Estimate for $H_j^{(n)}$

To bound $\|H_j^{(n)}\|$ we use the following standard result.

Lemma 1. Let $-\overline{A}$ be a Hurwitz matrix. Then for any $P = P^\top \succ 0$ there exists a unique $Q = Q^\top \succ 0$ satisfying

$$
\overline{A}^\top Q + Q\overline{A} = P.
$$

Moreover, letting

$$
a = \frac{\lambda_{\min}(P)}{2\,\|Q\|}, \quad \alpha_\infty = \frac{\lambda_{\min}(P)}{2\kappa_Q\,\|\overline{A}\|_Q^2} \wedge \frac{\|Q\|}{\lambda_{\min}(P)},
$$

with $\kappa_Q = \lambda_{\max}(Q)/\lambda_{\min}(Q)$, one has for all $\alpha \in [0, \alpha_\infty]$:

$$
\alpha a \le 1/2, \quad \|I - \alpha\overline{A}\|_Q^2 \le 1 - \alpha a.
$$

$$
\begin{aligned}
\|H_j^{(n)}\| &= \left\| \sum_{i=1}^{n-j} \big(I - \alpha\overline{A}\big)^{i-1} \big(I - 2\alpha\overline{A}\big)^{n-j-i} \right\| \\
&\le \sum_{i=1}^{n-j} \|I - \alpha\overline{A}\|^{i-1} \, \|I - 2\alpha\overline{A}\|^{n-j-i} \\
&\le \sum_{i=1}^{n-j} C_A (1-\alpha a)^{\frac{i-1}{2}} (1-2\alpha a)^{\frac{n-j-i}{2}} \\
&= C_A \sum_{k=0}^{n-j-1} (1-2\alpha a)^{k/2} (1-\alpha a)^{(n-j-k-1)/2} \\
&= C_A (1-\alpha a)^{(n-j-1)/2} \sum_{k=0}^{n-j-1} \left(\frac{1-2\alpha a}{1-\alpha a} \right)^{k/2} \\
&\le C_A (1-\alpha a)^{(n-j-1)/2} \frac{1}{1 - \left(\frac{1-2\alpha a}{1-\alpha a}\right)^{1/2}}
\end{aligned}
$$

$$
\left(\frac{1-2\alpha a}{1-\alpha a} \right)^{1/2} = \left(1 - \frac{\alpha a}{1-\alpha a} \right)^{1/2} = 1 - \frac{\alpha a}{2(1-\alpha a)} + O\left(\left(\frac{\alpha a}{1-\alpha a} \right)^2 \right) = 1 - \frac{1}{2}\alpha a + O(\alpha^2)
$$

$$
1 - \left(\frac{1-2\alpha a}{1-\alpha a} \right)^{1/2} = \frac{1}{2}\alpha a + O(\alpha^2)
$$

$$\frac{1}{1-\left(\frac{1-2\alpha a}{1-\alpha a}\right)^{1/2}} = \frac{2}{\alpha a} + O(1)$$

$$\|H_j^{(n)}\| \leq \overline{C}_A (1-\alpha a)^{(n-j-1)/2} \frac{2}{\alpha a}$$

To explore $\tilde{J}_n^{(0,\alpha)}$ we use the following statement:

Lemma 2. Assume **UGE 1**. Let $\{g_i\}_{i=1}^n$ be a family of measurable functions $g_i : Z \to \mathbb{R}^d$ such that

$$c_i = \|g_i\|_\infty < \infty \quad \text{for all } i \geq 1, \quad \pi(g_i) = 0 \quad \text{for all } i \in \{1, ..., n\}.$$

Then, for any initial distribution ξ on $(Z, \mathcal{Z})$, any $n \in \mathbb{N}$ and any $t \geq 0$,

$$\mathbb{P}_\xi\left(\left\|\sum_{i=1}^n g_i(Z_i)\right\| \geq t\right) \leq 2\exp\left(-\frac{t^2}{2u_n^2}\right),$$

where

$$u_n = 8\left(\sum_{i=1}^n c_i^2\right)^{1/2} \sqrt{t_{\text{mix}}}.$$

Proof. The proof follows the lines of Durmus et al. (2025, Lemma 9).

$$\tilde{J}_n^{(0,\alpha)} = -\sum_{j=1}^n 2\alpha^2 \overline{A} H_j^{(n)} \epsilon(Z_j)$$

$$\|2\alpha^2 \overline{A} H_j^{(n)}\|_\infty \leq 2\alpha^2 \tilde{C}_A (1-\alpha a)^{(n-j-1)/2} \frac{2}{\alpha a} = 4\alpha \tilde{C}_A (1-\alpha a)^{(n-j-1)/2}$$

$$\text{where} \quad \tilde{C}_A = \|A\| \, \overline{C}_A$$

$$\begin{aligned} u_n^2 &\leq 64 \cdot 16\alpha^2 \tilde{C}_A^2 \sum_{j=1}^n (1-\alpha a)^{n-j-1} \\ &= \hat{C}_A^2 \alpha^2 \frac{1}{1-\alpha a} \sum_{k=0}^{n-1} (1-\alpha a)^k \\ &\leq \hat{C}_A^2 \alpha^2 \frac{1}{1-\alpha a} \frac{1}{1-1+\alpha a} \\ &= \hat{C}_A^2 \alpha \end{aligned}$$

Now use classic lemma:

Lemma 3. Let X be an $\mathbb{R}^d$-valued random variable satisfying

$$\mathbb{P}(\|X\| \geq t) \leq 2\exp\left(-\frac{t^2}{2\sigma^2}\right) \quad \text{for all } t \geq 0,$$

for some $\sigma^2 > 0$. Then, for any $p \geq 2$, it holds that

$$\mathbb{E}[\![|X\|^p] \leq 2\, p^{p/2}\, \sigma^p.$$

$$\frac{1}{\sqrt{\alpha}} \mathbb{E}^{1/2} \, \|\tilde{J}_n^{(0,\alpha)}\|^2 \leq \frac{1}{\sqrt{\alpha}} \cdot 2 \cdot \sqrt{\alpha} \hat{C}_A = 2\hat{C}_A$$

3. New Lemma

3.1. Study Asymptotic of $J_n^{(1,\alpha)}$

Let's remember some formulas:

$$J_n^{(0,\alpha)} = \left(I - \alpha\overline{A}\right)J_{n-1}^{(0,\alpha)} - \alpha\epsilon(Z_n), \quad J_0^{(0,\alpha)} = 0$$

$$J_n^{(1,\alpha)} = \left(I - \alpha\overline{A}\right)J_{n-1}^{(1,\alpha)} - \alpha\tilde{A}(Z_n)J_{n-1}^{(0,\alpha)}$$

$$J_n^{(1,\alpha)} = -\alpha\sum_{j=1}^{n}\left(I - \alpha\overline{A}\right)^{n-j}\tilde{A}(Z_j)$$

$$J_n^{(1,\alpha)} = -\alpha^2\sum_{j=1}^{n}\sum_{l=1}^{j-1}\left(I - \alpha\overline{A}\right)^{n-j}\tilde{A}(Z_j)\left(I - \alpha\overline{A}\right)^{j-1-l}\epsilon(Z_l)$$

$Y_t = \left(J_t^{(0,\alpha)}, Z_{t+1}\right)$ is a Markov Chain with $\Pi_{J,\alpha}$ invariant distribution.

Let $\psi(J,z) = \tilde{A}(z)J$:

$$\overline{\psi}_t = \psi\left(J_t^{(0,\alpha)}, Z_{t+1}\right) - \mathbb{E}_{\Pi_{J,\alpha}}[\psi_0]$$

We know that:

$$\left(\mathbb{E}_{\Pi_{J,\alpha}}\left\|\sum_{t=0}^{n-1}\overline{\psi}_t\right\|^p\right)^{1/p} \lesssim C_A\kappa_Q^{1/2}\rho^{1/2}t_{\text{mix}}^{1/2}\left\|\epsilon\right\|_\infty\left(pa^{-1/2}(\alpha n)^{1/2} + t_{\text{mix}}^{1/2}\alpha n^{1/2} + a^{-1/2}\alpha^{1/2}\right) \lesssim C\sqrt{\alpha n}$$

Let's study:

$$\zeta_t(J,z) = \left(I - \alpha\overline{A}\right)^{n-t}\tilde{A}(Z_{t+1})J_t^{(0,\alpha)}$$

and:

$$\left(\mathbb{E}_{\Pi_{J,\alpha}}\left\|\sum_{t=0}^{n-1}\overline{\zeta}_t\right\|^p\right)^{1/p}$$

$$B^k = (I - \alpha A)^k \quad \|B^k\|_Q \le (1-\alpha a)^{k/2} \Longrightarrow \sum_{k=0}^{n-1}\|B^k\|_Q^2 \le \sum_{k=0}^{n-1}(1-\alpha a)^k \le \frac{1}{\alpha a}$$

$$\zeta_t = B^{n-t}\tilde{A}(Z_{t+1})J_t^{(0,\alpha)} = -\alpha B^{n-t}\tilde{A}(Z_{t+1})\left(\sum_{k=1}^{t}B^{t-k}\epsilon(Z_k)\right) = -\alpha\sum_{k=1}^{t}B^{n-t}\tilde{A}(Z_{t+1})B^{t-k}\epsilon(Z_k)$$

$$\begin{aligned}
S_n &= -\alpha\sum_{t=1}^{n-1}\sum_{k=1}^{t}B^{n-t}\tilde{A}(Z_{t+1})B^{t-k}\epsilon(Z_k) \\
&= -\alpha\sum_{k=1}^{n-1}\left(\sum_{l=1}^{n-k}B^{n-(k+l-1)}\tilde{A}(Z_{t+1})B^{l-1}\right)\epsilon(Z_k) \\
&= -\alpha\sum_{k=1}^{n-1}H_{k+1}^{(w)}\epsilon(Z_k)
\end{aligned}$$

$$H_{k+1}^{(w)} = \sum_{l=1}^{n-k} B^{n-(k+l-1)} \tilde{A}(Z_{k+l}) B^{l-1}$$

$$\mu_{t,k}^{(w)} = \mathbb{E}_\pi \left[B^{n-t} \tilde{A}(Z_{t+1}) B^{t-k} \epsilon(Z_k) \right]$$

$$\mu_k^{(w)} = \mathbb{E}_\pi \left[\sum_{l=1}^{n-k} B^{n-(k+l-1)} \tilde{A}(Z_{k+l}) B^{l-1} \epsilon(Z_k) \right]$$

$$S_n - \mathbb{E} S_n = -\alpha \sum_{k=1}^{n-1} \left(H_{k+1}^{(w)} \epsilon(Z_k) - \mu_k^{(w)} \right)$$

$$H_{k+1}^{(w)} \epsilon(Z_k) - \mu_k^{(w)} = H_{k+1}^{(w)} \epsilon(Z_k) - \mathbb{E}\left[H_{k+1}^{(w)} \epsilon(Z_k) \mid \mathcal{F} \right] + \mathbb{E}\left[H_{k+1}^{(w)} \epsilon(Z_k) \mid \mathcal{F} \right] - \mu_k^{(w)}$$
$$\text{where} \quad \mathcal{F} = \sigma(Z_1, ..., Z_k)$$

Decompose it to martingale + residual:

$$M_k^{(w)} = H_{k+1}^{(w)} \epsilon(Z_k) - \mathbb{E}\left[H_{k+1}^{(w)} \epsilon(Z_k) \mid \mathcal{F} \right]$$

$$R_k^{(w)} = \mathbb{E}\left[H_{k+1}^{(w)} \epsilon(Z_k) \mid \mathcal{F} \right] - \mu_k^{(w)}$$

$$U_1^{(w)} = H_2^{(W)} \epsilon(Z_1) - \mu_1^{(w)} \quad U_2^{(w)} = \sum_{k=2}^{n-1} M_k^{(w)} \quad U_3^{(w)} = \sum_{k=2}^{n-1} R_k^{(w)}$$

$$S_n - \mathbb{E} S_n = -\alpha \left(U_1^{(w)} + U_2^{(w)} + U_3^{(w)} \right) \quad \|\epsilon(Z)\| \le \|\epsilon\|_\infty$$

$$\|M_k^{(w)}\|_{L_p} \le 2 \, \|H_{k+1}^{(w)} \epsilon(Z_k)\|_{L_p} \le 2 \, \|\epsilon\|_\infty \sup_{\|u\|=1} \|H_{k+1}^{(w)} u\|_{L_p}$$

$$H_{k+1}^{(w)} u = \sum_{l=1}^{n-k} g_{k,l}(Z_{k+l}) \quad g_{k,l}(Z) = B^{n-(n-k-1)} \tilde{A}(Z) B^{l-1} u$$

Lemma 4. Assume **UGE 1**. Let $\{g_i\}_{i=1}^n$ be a family of measurable functions $g_i : Z \to \mathbb{R}^d$ such that

$$c_i = \|g_i\|_\infty < \infty \quad \text{for all } i \ge 1, \quad \pi(g_i) = 0 \quad \text{for all } i \in \{1, ..., n\}.$$

Then, for any initial distribution ξ on $(Z, \mathcal{Z})$, any $n \in \mathbb{N}$ and any $t \ge 0$,

$$\mathbb{P}_\xi \left(\left\| \sum_{i=1}^n g_i(Z_i) \right\| \ge t \right) \le 2 \exp\left(-\frac{t^2}{2u_n^2} \right),$$

where

$$u_n = 8 \left(\sum_{i=1}^n c_i^2 \right)^{1/2} \sqrt{t_{\text{mix}}}.$$

Proof. The proof follows the lines of Durmus et al. (2025, Lemma 9).

$$c_{k,l} = \|g_{k,l}\|_\infty \quad u_k = 8\sqrt{t_{\text{mix}}} \left(\sum_{l=1}^{n-k} c_{k,l}^2 \right)^{1/2} \implies \|H_{k+1}^{(w)} u - \mathbb{E}\left[H_{k+1}^{(w)} u \right]\|_{L_p} \le p^{1/2} u_k$$

$$c_{k,l} = \sup_Z \|B^{n-(n-k-1)}\tilde{A}(Z)B^{l-1}u\| \le C_A \cdot \|B^{n-k-l+1}\| \cdot \|B^{l-1}u\|$$

$$\sum_{l=1}^{n-k} c_{k,l} \le C^2 \sum_{l=1}^{n-k} (1-\alpha a)^{n-k-l+1+l-1} = C^2(n-k)(1-\alpha a)^{n-k}$$

$$\|H_{k+1}^{(w)}u - \mathbb{E}\Big[H_{k+1}^{(w)}u\Big]\|_{L_p} \le Cp^{1/2}t_{\text{mix}}^{1/2}\sqrt{(n-k)(1-\alpha a)^{n-k}}$$

$$\begin{aligned}
\|U_2^{(w)}\|_{L_p} &\le p\left(\sum_{k=2}^{n-1}\|M_k^{(w)}\|_{L_p}^2\right)^{1/2} \\
&\lesssim Cp^{3/2}t_{\text{mix}}^{1/2}\ \|\epsilon\|_\infty\ \left(\sum_{k=2}^{n-1}(n-k)(1-\alpha a)^{n-k}\right)^{1/2} \\
&\lesssim Cp^{3/2}t_{\text{mix}}^{1/2}\ \|\epsilon\|_\infty\ \frac{1}{\alpha a}
\end{aligned}$$

$$v_k^{(w)}(z) = \sum_{l=1}^{n-k} Q^l\left[B^{n-k-l+1}\tilde{A}(z)\right]B^{l-1}, \quad v_k^{(w,\epsilon)}(z) = v_k^{(w)}(z)\epsilon(z)$$

$$\mathbb{E}\Big[H_{k+1}^{(w)}\epsilon(Z_k) \mid \mathcal{F}_k\Big] = \left(\sum_{l=1}^{n-k} B^{n-k-l+1}\Big(Q^l\tilde{A}\Big)(z)B^{l-1}\right)\epsilon(Z_k) = v_k^{(w,\epsilon)}(Z_k)$$

$$\mathbb{E}\Big[H_{k+1}^{(w)}\epsilon(Z_k) \mid \mathcal{F}_{k-1}\Big] = \Big(Qv_k^{(w,\epsilon)}\Big)(Z_{k-1}) = \varphi_k(Z_{k-1}) = Qg_k$$

$$\overline{\varphi}_k(z) = \varphi_k(z) - \mu_k^{(w)}$$

$$U_3^{(w)} = \sum_{k=2}^{n-1}\overline{\varphi}_k(Z_{k-1})$$

$$\|\overline{\varphi}_k\|_\infty \le \sup_{z,z'}\|\varphi_k(z) - \varphi_k(z')\| \le 2\Delta(Q)\ \|g\|_\infty \le 2\ \|g\|_\infty \le 2\ \|v_k^{(w)}\|_\infty\ \|\epsilon\|_\infty$$

$$\|v_k^{(w)}\|_\infty \le \sum_{l=1}^{n-k}\|B^{n-k-l+1}\|_Q\ \|\Big(Q^l\tilde{A}\Big)(\cdot)\|_\infty\ \|B^{l-1}\|_Q$$

Using $\pi\Big(\tilde{A}\Big) = 0$ and the definition of the Dobrushin coefficient,

$$\|\Big(Q^l\tilde{A}\Big)(z)\| = \left\|\int \tilde{A}(u)\big(Q^l(z,du) - \pi(du)\big)\right\| \le 2C_A\Delta\big(Q^l\big).$$

Therefore,

$$\|v_k^{(w)}\|_\infty \le 2C_A\sum_{l=1}^{n-k}(1-\alpha a)^{(n-k-l+1)/2}(1-\alpha a)^{(l-1)/2}\Delta\big(Q^l\big) = 2C_A(1-\alpha a)^{(n-k)/2}\sum_{l=1}^{n-k}\Delta\big(Q^l\big).$$

By submultiplicativity of the Dobrushin coefficient, $\Delta\big(Q^l\big) \le \Delta(Q)^l$, hence

$$\sum_{l=1}^{n-k}\Delta\big(Q^l\big) \le \sum_{l=1}^{\infty}\Delta(Q)^l = \frac{\Delta(Q)}{1-\Delta(Q)} \lesssim t_{\text{mix}}.$$

Consequently,

$$\|\overline{\varphi}_k\|_\infty \le C(1-\alpha a)^{(n-k)/2} t_{\text{mix}} \, \|\epsilon\|_\infty.$$

Applying Lemma 11 to $U_3^{(w)} = \sum_{k=2}^{n-1} \overline{\varphi}_k(Z_{k-1})$, we obtain

$$\|U_3^{(w)}\|_{L_p} \lesssim p^{1/2}\sqrt{t_{\text{mix}}}\left(\sum_{k=2}^{n-1}\|\overline{\varphi}_k\|_\infty^2\right)^{1/2} \lesssim p^{1/2} t_{\text{mix}} \, \|\epsilon\|_\infty \, \left(\sum_{k=2}^{n-1}(1-\alpha a)^{n-k}\right)^{1/2}.$$

Since

$$\sum_{k=2}^{n-1}(1-\alpha a)^{n-k} = \sum_{j=1}^{n-2}(1-\alpha a)^j \le \frac{1}{\alpha a},$$

we conclude that

$$\|U_3^{(w)}\|_{L_p} \lesssim p^{1/2} t_{\text{mix}} \, \|\epsilon\|_\infty \, \frac{1}{\sqrt{\alpha a}}.$$

Combining the bounds for $U_1^{(w)}, U_2^{(w)}, U_3^{(w)}$, we finally obtain

$$\left(\mathbb{E}_{\Pi_{J,\alpha}} \, \|S_n - \mathbb{E}S_n\|^p\right)^{1/p} \lesssim \alpha\left(p^{3/2} t_{\text{mix}}^{1/2}\frac{1}{\alpha a} + p^{1/2} t_{\text{mix}}\frac{1}{\sqrt{\alpha a}}\right)$$